Multi-Factor Intelligent Major Recommendation System

Decision Support Using Logical, Emotional, and Psychological Analysis

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PROBLEM & OBJECTIVE

"Students often choose majors based on biased or incomplete information, leading to academic mismatch and dissatisfaction."

"**Goal:** Build an AI-driven system providing personalized, explainable recommendations based on holistic student profiles."

KEY RESULTS

Top 3 Rankings
Majors ranked with compatibility percentages

Transparent Reasoning
Step-by-step explanation for top recommendation

Validated Inputs
Handles conflicting/extreme profiles robustly


Explanation

Factor Contribution to Score

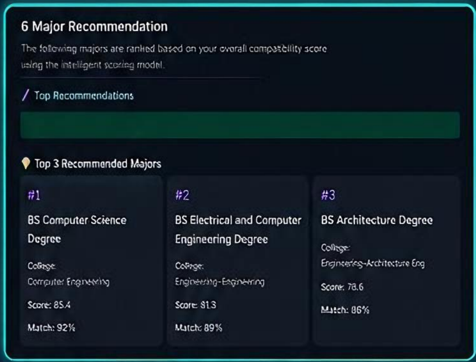
Factor	BS Computer Science Degree	BS Electrical and Computer Engineering Degree	BS Architecture Degree
1. Logical Score (Grades)	34.0	34.0	32.0
2. Emotional Score (Interests)	27.0	24.0	21.0
3. Psychological Score (MBTI)	14.0	15.0	13.0
4. Market Score (Job Trends)	0.0	8.0	7.0
Total Weighted Score	85.0	81.0	71.0
Compatibility Percentage	92 %	89 %	[illegible]
Recommendation Rank	1	2	3

"Explainability: Reasoning shown for every recommendation"

INTELLIGENT SCORING MODEL

$$\text{Final Score} = 0.40 \times \text{Logical} + 0.30 \times \text{Emotional} + 0.20 \times \text{Psychological} + 0.10 \times \text{Market}$$

- Logical (40%) – Grades & Academic Skills
- Emotional (30%) – Interests & Passions
- Psychological (20%) – Personality (MBTI)
- Market (10%) – Job Demand Trends



"System output: Top 3 ranked majors with compatibility scores"

IMPLEMENTATION

- Python
- Streamlit
- Interactive UI
- Real-time Scoring
- Explanation Module
- Robustness Validation

IMPACITION & FUTURE ROADMAP

- Enhances decision-making transparency and student satisfaction
- **Next:** ML model integration, real-world datasets, cloud deployment